David Bani-Harouni

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dharouni Google Scholar

Education

Technical University of Munich — Munich, Germany
PhD in Computer Science — June 2023 – present
- Supervisor: Prof. Nassir Navab
- Focus: Reinforcement learning for large language models
- Published first-author works at ICLR, MICCAI, and Nature Scientific Reports

Technical University of Munich — Munich, Germany
MSc in Biomedical Computing — Oct 2020 – Jan 2023
- GPA: 1.3 (top 2%)
- Thesis: Deep Learning with Individual Differential Privacy (published at MICCAI CareAI 2023, paper), Prof. Daniel Rückert

Technical University of Munich — Munich, Germany
BSc in Informatics — Oct 2016 – Aug 2020
- GPA: 1.7
- Minor: Mathematics
- Thesis: Machine Learning based Clinical Decision Support System for Poison Diagnosis (published at MICCAI 2020, paper), Prof. Nassir Navab
- Exchange semester: Computer Science at *National University of Singapore*, Aug 2018 – Dec 2018

Experience

Student Research Assistant — Munich, Germany
Technical University of Munich — June 2020 – Aug 2021
- Worked at the Computer Aided Medical Procedures group of Prof. Nassir Navab
- Developed a graph-based artificial intelligence architecture to support clinicians in the outcome prediction of patients suffering from a COVID-19 infection
- Co-first-authored journal paper published at Nature Scientific Reports

Teaching Assistant — Munich, Germany
Technical University of Munich — Apr 2018 – Aug 2018
- Taught *Introduction to Software Engineering*

Software Development Intern — Hamburg, Germany
inovex — Aug 2017 – Oct 2017
- Supported the development of a Java-based robotics platform

Key Publications

Rewarding Doubt: A Reinforcement Learning Approach to Calibrated Confidence Expression of Large Language Models — 2026
David Bani-Harouni*, Chantal Pellegrini*, Paul Stangel, Ege Özsoy, Kamilia Zaripova, Matthias Keicher, Nassir Navab
ICLR
arxiv.org/abs/2503.02623

Calibrated Confidence Expression for Radiology Report Generation 2026
David Bani-Harouni*, Chantal Pellegrini*, Julian Lüers, Su Hwan Kim, Markus Baalmann, Benedikt Wiestler, Rickmer Braren, Nassir Navab, Matthias Keicher
MICCAI
arxiv.org/abs/2603.29492

Language Agents for Hypothesis-driven Clinical Decision Making with Reinforcement Learning 2026
David Bani-Harouni, Chantal Pellegrini, Ege Özsoy, Matthias Keicher, Nassir Navab
ICLR
arxiv.org/abs/2506.13474

MAGDA: Multi-agent guideline-driven diagnostic assistance 2024
David Bani-Harouni, Nassir Navab, Matthias Keicher
MICCAI MedAGI
arxiv.org/abs/2409.06351

Awards

best.in.tum: Program for top 2% of students of the Informatics department at TUM

Deutschlandstipendium: Scholarship awarded by TUM

DAAD RISE Worldwide 2020: Scholarship for a research project abroad (canceled due to the COVID-19 pandemic)

Skills

Programming: Python, Java, C++, SQL

ML Frameworks: PyTorch, Hugging Face

Tools & Platforms: Slurm, Linux, Git, LaTeX

Research Areas: Natural Language Processing, Medical Imaging, Reinforcement Learning

Languages: English (fluent), German (native)

All Publications

Calibrated Confidence Expression for Radiology Report Generation 2026
David Bani-Harouni*, Chantal Pellegrini*, Julian Lüers, Su Hwan Kim, Markus Baalmann, Benedikt Wiestler, Rickmer Braren, Nassir Navab, Matthias Keicher
MICCAI
arxiv.org/abs/2603.29492

Learning Diagnostic Reasoning for Decision Support in Toxicology 2026
Nico Oberländer*, **David Bani-Harouni***, Tobias Zellner, Nassir Navab, Florian Eyer, Matthias Keicher
MICCAI AMAI (Best Student Paper Award)
arxiv.org/abs/2603.29608

Speechct-clip: Distilling text-image knowledge to speech for voice-native multimodal ct analysis 2026
Lukas Buess*, Jan Geier*, **David Bani-Harouni**, Chantal Pellegrini, Matthias Keicher, Paula Andrea Perez-Toro, Nassir Navab, Andreas Maier, Tomas Arias-Vergara
ICASSP
arxiv.org/abs/2510.02322

Specialized foundation models for intelligent operating rooms 2026
Ege Özsoy, Chantal Pellegrini, **David Bani-Harouni**, Kun Yuan, Matthias Keicher, Nassir Navab
npj Digital Medicine
www.nature.com/articles/s41746-026-02631-4

From EHRs to Patient Pathways: Scalable Modeling of Longitudinal Health Trajectories with LLMs 2026
Chantal Pellegrini, Ege Özsoy, **David Bani-Harouni**, Matthias Keicher, Nassir Navab
TMLR
arxiv.org/abs/2506.04831

Language Agents for Hypothesis-driven Clinical Decision Making with Reinforcement Learning 2026
David Bani-Harouni, Chantal Pellegrini, Ege Özsoy, Matthias Keicher, Nassir Navab
ICLR
arxiv.org/abs/2506.13474

Rewarding Doubt: A Reinforcement Learning Approach to Calibrated Confidence Expression of Large Language Models 2026
David Bani-Harouni*, Chantal Pellegrini*, Paul Stangel, Ege Özsoy, Kamilia Zaripova, Matthias Keicher, Nassir Navab
ICLR
arxiv.org/abs/2503.02623

Knowledge Graph Sparsification for GNN-based Rare Disease Diagnosis 2025
Premt Cara, Kamilia Zaripova, **David Bani-Harouni**, Nassir Navab, Azade Farshad
Preprint
arxiv.org/abs/2510.08655

Mm-or: A large multimodal operating room dataset for semantic understanding of high-intensity surgical environments 2025
Ege Özsoy, Chantal Pellegrini, Tobias Czempiel, Felix Tristram, Kun Yuan, **David Bani-Harouni**, Ulrich Eck, Benjamin Busam, Matthias Keicher, Nassir Navab
CVPR
arxiv.org/abs/2503.02579

MAGDA: Multi-agent guideline-driven diagnostic assistance 2024
David Bani-Harouni, Nassir Navab, Matthias Keicher
MICCAI MedAGI
arxiv.org/abs/2409.06351

Multimodal graph attention network for COVID-19 outcome prediction 2023
Matthias Keicher*, Hendrik Burwinkel*, **David Bani-Harouni***, Magdalini Paschali, Tobias Czempiel, Egon Burian, Marcus R. Makowski, Rickmer Braren, Nassir Navab, Thomas Wendler
Nature Scientific Reports
arxiv.org/abs/2108.00860

Gradient Self-alignment in Private Deep Learning 2023
David Bani-Harouni, Tamara T. Mueller, Daniel Rueckert, Georgios Kaissis
MICCAI CareAI

ToxNet: an artificial intelligence designed for decision support for toxin prediction 2023
Tobias Zellner, Katrin Romanek, Christian Rabe, Sabrina Schmoll, Stefanie Geith, Eva-Carina Heier, Raphael Stich, Hendrik Burwinkel, Matthias Keicher, **David Bani-Harouni**, Nassir Navab, Seyed-Ahmad Ahmadi, Florian Eyer
Clinical Toxicology

Decision support for intoxication prediction using graph convolutional networks 2020
Hendrik Burwinkel*, Matthias Keicher*, **David Bani-Harouni**, Tobias Zellner, Nassir Navab, Seyed-Ahmad Ahmadi, Florian Eyer
MICCAI
arxiv.org/abs/2005.00840